

# CS & IT ENGINEERING

Digital Logic

MISCALLENIOUS

Lecture No. 3



By- Aditya sir

# Recap of Previous Lecture



## Topic



→ Number Systems

→ 1's Compl Rep $\times$

→ 2's Compl Rep $\times$

→ Signed/Unsigned Nos



# Topics to be Covered



## Topic



- Kmap with a variable
- Design of a Synchronous Counter
- Misc Questions



## About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.







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# [MCQ]



V. Imp ★

#Q. Find the base values which can satisfy the following equation.

$$\left(\frac{39}{3} = 13\right)_x$$

27%

**A** Only 10 ✗

**B** Any base ✗

**C** Any base > 9

**D** Any base  $\geq 9$  ✗

13  $\rightarrow$  (D)

13

Soln:-  $\left(\frac{39}{3} = 13\right)_x$

$$\left(\frac{(39)_x}{(3)_x} = (13)_x\right)$$

$$\frac{3 \times x^1 + 9 \times x^0}{3 \times x^0} = 1 \times x^1 + 3 \times x^0$$

$$\frac{3x+9}{3} = x+3 \Rightarrow \boxed{x+3 = x+3} \rightarrow \text{for all values of } x.$$

But  $\begin{matrix} (39)_x \\ \downarrow \end{matrix} \Rightarrow x > 3 \text{ \& } x > 9 \text{ (then only valid)}$

Hence any  $\boxed{x > 9} \Rightarrow \underline{\text{Base} > 9}$

Variable inside the K-map



Q1:-

$f(A, B, C)$

A Karnaugh map for the function  $f(A, B, C)$ . The map is a 2x2 grid with columns labeled  $\overline{B}$  and  $B$ , and rows labeled  $\overline{A}$  and  $A$ . The cells contain values for  $C$ : the top-left cell (0) has  $C=0$ , the top-right cell (1) has  $C=1$ , the bottom-left cell (2) is empty, and the bottom-right cell (3) is empty. Two yellow curved arrows group the two cells in the top row, indicating the expression  $\overline{A}(\overline{B} + B)$ .

|                |          |          |
|----------------|----------|----------|
| $\overline{A}$ | $C$<br>0 | $C$<br>1 |
| $A$            |          |          |
|                | 2        | 3        |

$$\Rightarrow \overline{A}\overline{B} \cdot C + \overline{A}B \cdot C$$

$$= \overline{A}C(\overline{B} + B)$$

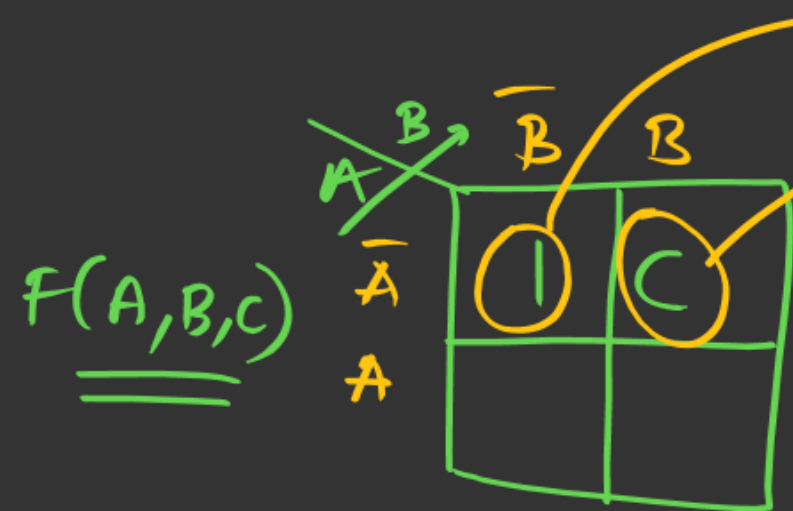
$$= \boxed{\overline{A}C} \checkmark$$

A Karnaugh map for the function  $f(A, B)$ . The map is a 2x2 grid with columns labeled  $\overline{B}$  and  $B$ , and rows labeled  $\overline{A}$  and  $A$ . The cells contain values: the top-left cell has 1, the top-right cell has 1, the bottom-left cell has 0, and the bottom-right cell has 0. A diagonal line is drawn from the top-left to the bottom-right, representing the expression  $\overline{A}\overline{B}$ .

|                |   |   |
|----------------|---|---|
| $\overline{A}$ | 1 | 1 |
| $A$            | 0 | 0 |

$$= \overline{A}\overline{B} \textcircled{1}$$

eg2 :- Simplify:-



Appx1:-

$$\Rightarrow \bar{A}\bar{B} \cdot 1 + \bar{A}BC$$

$$= \bar{A}(\bar{B} + BC)$$

$$= \bar{A}((\bar{B} + B)(\bar{B} + C))$$

$$= \bar{A}(\bar{B} + C) \checkmark$$

$$= \boxed{\bar{A}\bar{B} + \bar{A}C} \checkmark$$

Ques:- Simplify:-

$f(A, B, C)$

|              |              |   |
|--------------|--------------|---|
| <del>A</del> | <del>B</del> |   |
|              | 1            | C |
|              |              |   |

App<sup>o</sup> 2:- Case 1:-  $C = 0$

$\Rightarrow$

|           |   |           |   |
|-----------|---|-----------|---|
|           | B | $\bar{B}$ | B |
| A         |   | 1         | 0 |
| $\bar{A}$ |   | 0         | 0 |
| A         |   | 0         | 0 |

$\Rightarrow \bar{A} \bar{B} \cdot \bar{C}$

Case 2:-  $C = 1$   
 $\Rightarrow$

|           |   |           |   |
|-----------|---|-----------|---|
|           | B | $\bar{B}$ | B |
| A         |   | 1         | 1 |
| $\bar{A}$ |   | 0         | 0 |
| A         |   | 0         | 0 |

$\Rightarrow \bar{A} \cdot C$

$$f(A, B, C) = \bar{A} \bar{B} \bar{C} + \bar{A} C$$

$$= \bar{A} (\bar{B} \bar{C} + C)$$

$$= \bar{A} ((\bar{B} + C)(C + \bar{C}))$$

$$= \underline{\bar{A} \bar{B} + \bar{A} C}$$



Type-2:

Simplify

$$F(A, B, C) = \overline{A}\overline{B} + \overline{A}BC$$

✓ ① Boolean Algebra

✓ ② Standard K-map (minterm)

✓ ③ K-map with a variable.

Soln: ① Boolean Algebra:

$$F(A, B, C) = \bar{A}\bar{B} + \bar{A}BC$$

$$= \bar{A}(\bar{B} + BC)$$

$$= \bar{A}((\bar{B} + B)(\bar{B} + C)) = \boxed{\bar{A}\bar{B} + \bar{A}C} \checkmark$$

② Standard k-map :-

a) SOP

$$F(A, B, C) = \bar{A}\bar{B} + \bar{A}BC$$

$$= \bar{A}\bar{B}(C + \bar{C}) + \bar{A}BC$$

$$= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}BC$$

$$001, 000, 011$$

1      0      3

$$= \sum m(0, 1, 3)$$

| $\begin{array}{c} BC \\ \swarrow \searrow \\ A \end{array}$ | $\bar{B}\bar{C}$ | $\bar{B}C$ | $BC$ | $B\bar{C}$ |
|-------------------------------------------------------------|------------------|------------|------|------------|
| $\bar{A}$                                                   | 0                | 1          | 1    |            |
| $A$                                                         |                  |            |      |            |
|                                                             | 4                | 5          | 7    | 6          |

$\bar{A}\bar{B}$  points to cell 0,  $\bar{A}C$  points to cells 1 and 3.

$$= \bar{A}\bar{B} + \bar{A}C \checkmark$$



⑥ POS

$$F(A, B, C) = \overline{A}\overline{B} + \overline{A}BC$$

↓  
grouping: 0's

$$= \sum (0, 1, 3)$$

$$= \prod M(2, 4, 5, 6, 7)$$

POS

| $\begin{matrix} BC \\ \swarrow \nearrow \\ A \end{matrix}$ |   | $B+C$ | $B+\overline{C}$ | $\overline{B}+\overline{C}$ | $\overline{B}+C$ |
|------------------------------------------------------------|---|-------|------------------|-----------------------------|------------------|
|                                                            |   | 00    | 01               | 11                          | 10               |
| A                                                          | 0 | 0     | 1                | 3                           | 2                |
| $\overline{A}$                                             | 1 | 4     | 5                | 7                           | 6                |

$\overline{A}$  (points to row 1)  
 $\overline{B}+C$  (points to column 5)

$$= \overline{A} \cdot (\overline{B}+C)$$

$$= \overline{A}\overline{B} + \overline{A}C \checkmark$$

③ K-map with variable:

$$F(A,B,C) = \overline{A}\overline{B} + \overline{A}BC$$

$$= \underline{\overline{A}\overline{B}} \cdot 1 + \overline{A}B \cdot C$$

| $\begin{matrix} A & B \\ \hline \end{matrix}$ | $\overline{B}$ | $B$ |
|-----------------------------------------------|----------------|-----|
| $\overline{A}$                                | 1              | C   |
| $A$                                           |                |     |

$C=0$   
 $C=1$

|   |   |
|---|---|
| 1 | 0 |
| 0 | 0 |

$\overline{A}\overline{B}\overline{C}$

|   |   |
|---|---|
| 1 | 1 |
|   |   |

$\overline{A}C$

$$\begin{aligned} & \overline{A}\overline{B}\overline{C} + \overline{A}C \\ &= \overline{A}(\overline{B} + C) \end{aligned}$$

(8) Simplify:

|                |                |   |   |
|----------------|----------------|---|---|
| $\overline{A}$ | $\overline{B}$ | 1 | C |
| A              | B              |   | C |

Case 1:  $C=0$

|                |                |   |  |
|----------------|----------------|---|--|
| $\overline{A}$ | $\overline{B}$ | 1 |  |
| A              | B              |   |  |

$$\Rightarrow \overline{A} \overline{B} \overline{C}$$

Case 2:  $C=1$

|                |                |   |   |
|----------------|----------------|---|---|
| $\overline{A}$ | $\overline{B}$ | 1 | 1 |
| A              | B              | 1 | 1 |

$$\Rightarrow (\overline{A} + B) \cdot C$$

$$f = \overline{A} \overline{B} \overline{C} + (\overline{A} + B)C$$

$$= \overline{A} \overline{B} \overline{C} + \overline{A}C + BC$$

$$= \overline{A}(C + \overline{C} \overline{B}) + BC$$

$$= \overline{A}(\overline{B} + C) + BC$$

$$= \overline{A} \overline{B} + \overline{A}C + BC$$

010  
100

Redundancy then

$$= \underline{\overline{A} \overline{B} + BC} \checkmark$$

(Q.2) Simplify:

| A | B |   |
|---|---|---|
|   |   | 1 |
|   | C | C |

Case 1:  $C=0$

| $\bar{A}$ | $\bar{B}$ | B |                                     |
|-----------|-----------|---|-------------------------------------|
|           |           | 1 | $\rightarrow \bar{A}\bar{B}\bar{C}$ |
|           |           |   |                                     |

$C=1$

| $\bar{A}$ | $\bar{B}$ | B |          |
|-----------|-----------|---|----------|
|           |           | 1 | $(A+B)C$ |
| A         |           | 1 |          |

$$\hat{F} = \bar{A}\bar{B}\bar{C} + (A+B)C = \bar{A}\bar{B}\bar{C} + AC + BC$$

$$= B(\bar{A}\bar{C} + C) + AC$$

$$= B(\bar{A} + C) + AC = \bar{A}\bar{B} + \underbrace{BC + AC}_{\text{redundant}}$$

$$= \underline{\underline{\bar{A}B + AC}}$$



# Synchronous Counter Design

Step 1:- Write the previous and the Current State

Step 2:- Write Excitation table of FF.

Step 3:- Write logical Expression

Step 4:- Minimize

Step 5:- Hardware design

(Q.1) Design a Synchronous Counter using T FF which counts in the sequence:

$00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00 \rightarrow \dots$

00 → 11 → 01 → 10 → 00 ...

2 FFs

✓  
Step 1:-

Step 2

| prev  |       | curr    |         |       |       |
|-------|-------|---------|---------|-------|-------|
| $Q_1$ | $Q_0$ | $Q_1^+$ | $Q_0^+$ | $T_1$ | $T_0$ |
| 0     | 0     | 1       | 1       | 1     | 1     |
| 0     | 1     | 1       | 0       | 1     | 1     |
| 1     | 0     | 0       | 0       | 1     | 0     |
| 1     | 1     | 0       | 1       | 1     | 0     |

Step 3:- & Step 4

$$T_1 = 1$$

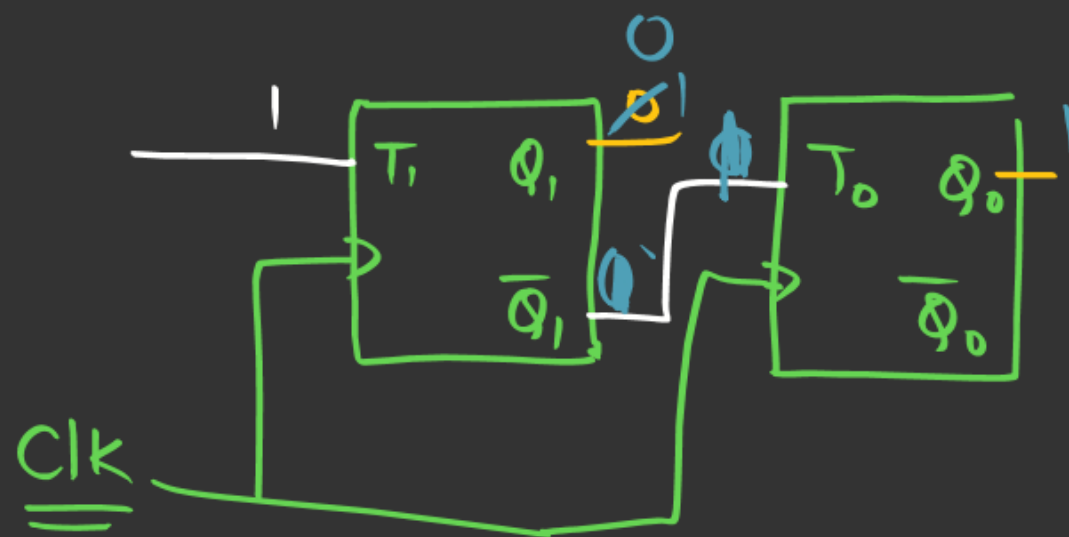
$$T_0 = \overline{Q_1} \overline{Q_0} + \overline{Q_1} Q_0$$

$$= \overline{Q_1} (\overline{Q_0} + Q_0)$$

$$\boxed{T_0 = \overline{Q_1}}$$



Steps: Hardware:



Validation: (Borno)

| CLK | $Q_1$ | $Q_0$ |
|-----|-------|-------|
| 0   | 0     | 0     |
| 1   | 1     | 1     |
| 2   | 0     | 1     |
| 3   | 1     | 0     |
| 4   | 0     | 0     |



(Q.2) Design a Synchronous Counter  
using D FF which counts

00  $\rightarrow$  11  $\rightarrow$  01  $\rightarrow$  00  $\rightarrow$  . . . . .

Imp obs:-

10  $\rightarrow$  unused  
State

10  $\rightarrow$  XX  
(dont care)

00 → 11 → 01 → 00 → . 2 DFFs

Step 1:-

10 → XX

prev curr Step 2

| $Q_1$ | $Q_0$ | $Q_1^+$ | $Q_0^+$ | $D_1$ | $D_0$ |
|-------|-------|---------|---------|-------|-------|
| 0     | 0     | 1       | 1       | 1     | 1     |
| 0     | 1     | 0       | 0       | 0     | 0     |
| ★ 1   | 0     | X       | X       | X     | X     |
| 1     | 1     | 0       | 1       | 0     | 1     |

Step 3 & Step 4:-

( $D_1$ )

|             |       |             |       |
|-------------|-------|-------------|-------|
|             | $Q_0$ | $\bar{Q}_0$ | $Q_0$ |
| $Q_1$       | 1     | 0           | 1     |
| $\bar{Q}_1$ | X     |             |       |
| $Q_1$       |       |             |       |

$$D_1 = \bar{Q}_0$$

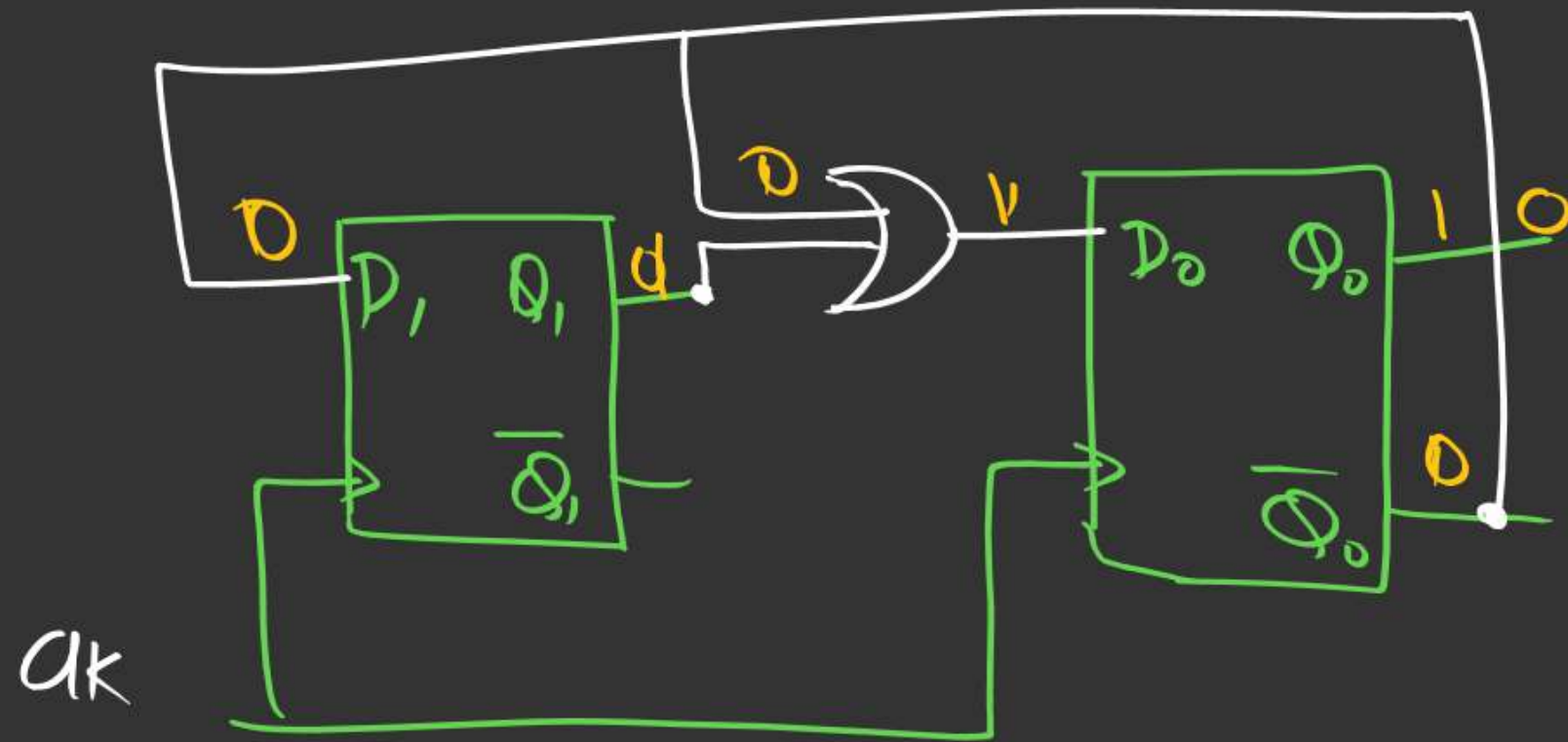
( $D_0$ )

|             |       |             |       |
|-------------|-------|-------------|-------|
|             | $Q_0$ | $\bar{Q}_0$ | $Q_0$ |
| $Q_1$       | 1     | 0           | 1     |
| $\bar{Q}_1$ | X     |             |       |
| $Q_1$       |       |             |       |

$$D_0 = \bar{Q}_0 + Q_1$$

$$D_1 = \overline{Q_0}$$

$$D_0 = Q_1 + \overline{Q_0}$$



Validation

| clk | Q <sub>1</sub> | Q <sub>0</sub> |
|-----|----------------|----------------|
| 0   | 0              | 0              |
| 1   | 1              | 1              |
| 2   | 0              | 1              |
| 3   |                |                |
| 4   |                |                |

## [MCQ]



#Q. Find the base values which can satisfy the following two equations simultaneously.

(i)  $2 + 3 = 5$

(ii)  $2 \times 4 = 10$

HW)

**A** Any base  $\geq 9$

**B** 8

**C** Any base  $\geq 8$

**D** 6



## [MCQ]



#Q. The result of addition operation  $34 + 43$  performed on minimum base is stored in an 8-bit register. The content of register will be

HW2

- A** 01000011
- B** 00101010
- C** 01010101
- D** 01010100



## 2 mins Summary



- k map with variable
- Design Sync Counter



**THANK - YOU**